

PHYS 4411: Physical Acoustics
UCA Student Success and Veterans Resource Center
DUE: Monday, April 27th, 2026 at 5:00PM

Background:

Give a brief background to the history of room acoustics, describe the Sabine equation, and its terms. What is reverberation time and why is it important? What are octave bands? Also describe the space being characterized (Mashburn 238). What are its dimensions and construction materials on ceiling, walls, and floor. What is the ideal reverberation time for this sized room? Some simplifying assumptions may be made if necessary.

Data Collection:

Highlight the software and method used to give the room an impulse and measure its response. What materials were used to try to modify the room's acoustic response? What were their surface areas and NRC values? Where were the materials placed in the room? Photographs help.

Theory & Analysis:

What software was used to determine the reverberation times in octave bands from the measured impulse response? What were the reverberation times for the empty room and the room with acoustic treatment in octave bands (show this data in tabular form)? What effect, if any, did the acoustic treatment have on the reverberation time in the octave bands measured?

Recommendations:

What recommendations can you make to the users of the room to improve the acoustics? What materials with a given NRC value and surface area could be used to improve the acoustics of the space? Estimate costs of these materials and note vendors. As a guide use the ANSI/ASA S12.60-2010 Part 1 standard for classroom acoustics. The relevant information is reproduced in Figure 1 for what reverberation time is appropriate for the space.

Report:

Write up a full report that includes your data, results of reverberation time analysis, and answers the questions in the sections above. Be sure to reference the ANSI/ASA standard when making your recommendations. Include any photographs, screen shots, code output, etc as needed to explain the data collection or analysis. You will work together as a team to write the report and do the analysis. Follow the IEEE format we have been using for reports.

Table 1 — Limits on A- and C-weighted sound levels of background noise and reverberation times in unoccupied furnished learning spaces

Learning space ^{a)}	Greatest one-hour average A- and C-weighted sound level of exterior-source background noise ^{b), f)} (dB)	Greatest one-hour average A- and C-weighted sound level of interior-source background noise ^{b), f)} (dB)	Maximum permitted reverberation times for sound pressure levels in octave bands with midband frequencies of 500, 1000, and 2000 Hz (s)
Core learning space with enclosed volume $\leq 283 \text{ m}^3$ ($\leq 10\,000 \text{ ft}^3$)	35 / 55	35 / 55	0.6 s ^{e)}
Core learning space with enclosed volume $> 283 \text{ m}^3$ and $\leq 566 \text{ m}^3$ ($> 10\,000 \text{ ft}^3$ and $\leq 20\,000 \text{ ft}^3$)	35 / 55	35 / 55	0.7 s
Core learning spaces with enclosed volumes $> 566 \text{ m}^3$ ($> 20\,000 \text{ ft}^3$) and all ancillary learning spaces	40 / 60 ^{d)}	40 / 60 ^{d)}	No requirement
a) See 3.1.1.1 and 3.1.1.2 for definitions of core and ancillary learning spaces. b) The greatest one-hour average A- and C-weighted interior-source and the greatest one-hour average A- and C-weighted exterior-source background noise levels are evaluated independently and will normally occur at different locations in the room and at different times of day. c) See 5.2.2 for other limits on interior-source background noise level. d) See 5.2.3 for limits in corridors adjacent to classrooms. e) See 5.3.2 for requirement that core learning spaces $\leq 283 \text{ m}^3$ ($\leq 10\,000 \text{ ft}^3$) shall be readily adaptable to allow reduction in reverberation time to 0.3 s. f) The design location shall be at a height of 1 m above the floor and no closer than 1 m from a wall, window, or fixed object such as HVAC equipment or supply or return opening. See A.1.3 for measurement location.			

Figure 1: Limits on A- and C-weighted sound levels of background noise and reverberation times in unoccupied furnished learning spaces from ANSI/ASA S12.60-2010 Part 1.